

A Semiparametric Approach to Density Estimation

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Abstract

- Proposes a semiparametric estimator that combines parametric and nonparametric density estimates.
- The approach adapts to the adequacy of the assumed parametric model.
- A mixing parameter determines the contribution of each estimator.
- The method provides both flexibility and efficiency.
- The mixing parameter also serves as a diagnostic for model fit.

Introduction

- **Parametric estimation:** Efficient when the model is correctly specified.
- **Nonparametric estimation:** Flexible but converges at a slower rate.
- Need for a unified framework that balances efficiency and robustness.
- The semiparametric approach integrates both methodologies.

Convex Combination of Densities

$$g(x, \pi) = \pi f(x, \hat{\theta}) + (1 - \pi) \hat{f}(x), \quad 0 \leq \pi \leq 1, \quad (1)$$

- $f(x, \hat{\theta})$: Parametric density with parameter estimated by maximum likelihood.
- $\hat{f}(x)$: Nonparametric kernel density estimator.
- π : Unknown mixing parameter estimated from the data.
- Interpretation:
 - $\pi \approx 1$ indicates a good parametric fit.
 - $\pi \approx 0$ favors the nonparametric estimate.

Nonparametric Density

$$\hat{f}(x) = c_0 \sum_{i=1}^n K[c_1(x - x_i)], \quad (2)$$

- $K(\cdot)$: Kernel function.
- c_1 : Bandwidth (smoothing parameter).
- c_0 : Normalizing constant ensuring integration to one.
- Provides flexibility without distributional assumptions.

Geometric Combination of CDFs

$$H(x, \pi) = F(x, \hat{\theta})^{\pi} \hat{F}(x)^{1-\pi}, \quad (3)$$

- $F(x, \hat{\theta})$: Parametric cumulative distribution function.
- $\hat{F}(x)$: Empirical or nonparametric CDF.
- Ensures a valid distribution function.
- Provides an alternative compromise between the two approaches.

Estimation of the Mixing Parameter

- The parameter $\hat{\theta}$ is obtained via maximum likelihood.
- The mixing parameter π is chosen to minimize the expected Hellinger distance between $g(x, \pi)$ and the true density.

$$H^2(f, g) = \frac{1}{2} \int \left(\sqrt{f(x)} - \sqrt{g(x, \pi)} \right)^2 dx. \quad (4)$$

- Asymptotic behavior:
 - If the parametric model is correct, $\pi \rightarrow 1$.
 - If the model is misspecified, $\pi \rightarrow 0$.

Numerical Results

Purpose of Numerical Examples

- Illustrate the practical performance of the semiparametric density estimator.
- Nonparametric component constructed using a **Gaussian kernel**.
- Bandwidth b is selected prior to estimating the mixing parameter π .
- Two datasets are analyzed:
 - Yearly maximum wind speed data.
 - Magnesium concentration in milk powder.

Convex Combination

$$g(x, \pi) = \pi f(x, \hat{\theta}) + (1 - \pi) \hat{f}(x), \quad 0 \leq \pi \leq 1. \quad (5)$$

- $f(x, \hat{\theta})$: Parametric density with parameters estimated by maximum likelihood.
- $\hat{f}(x)$: Kernel density estimator.
- π : Mixing parameter estimated numerically.
- Interpretation:
 - $\pi = 1$: Purely parametric estimator.
 - $\pi = 0$: Purely nonparametric estimator.
 - $0 < \pi < 1$: Compromise between the two.

Gaussian Kernel Density Estimator

Kernel Estimator

$$\hat{f}(x) = \frac{1}{nb} \sum_{i=1}^n K\left(\frac{x - x_i}{b}\right), \quad K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}. \quad (6)$$

- b : Bandwidth controlling the smoothness of $\hat{f}(x)$.
- A reasonable choice of b is assumed before estimating π .
- The semiparametric estimate tends to:
 - The nonparametric estimator as $b \rightarrow 0$.
 - The parametric estimator as $b \rightarrow \infty$.

Pseudolikelihood Function

$$\tilde{L}(\pi) = \prod_{i=1}^n \left[\pi f(x_i, \hat{\theta}) + (1 - \pi) \hat{f}(x_i) \right], \quad (7)$$

- The estimate $\hat{\pi}$ maximizes $\tilde{L}(\pi)$.
- Ensures stability and avoids bias from self-influence.

Example 1: Wind Speed Data

- Data: Yearly maximum wind speeds (North direction) from Sheridan, Wyoming.
- Sample size: $n = 20$ with observations:

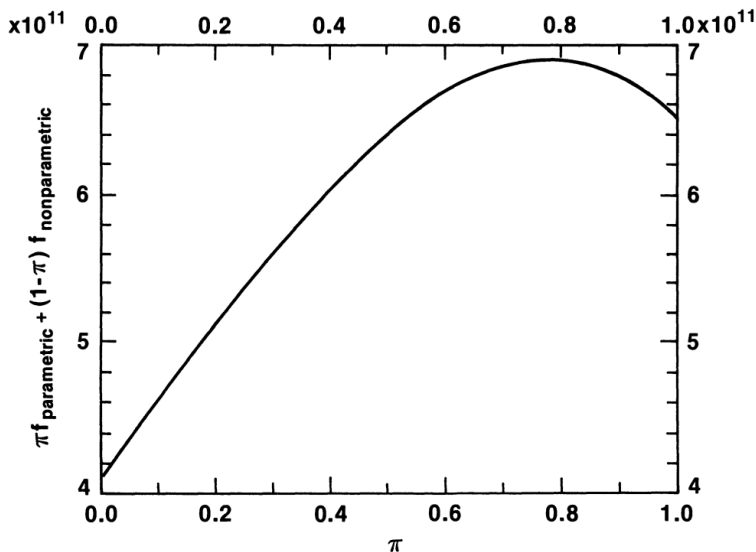
70, 61, 61, 60, 61, 63, 61, 67, 61, 62, 47, 67, 61, 49, 55, 65, 57, 51, 47, 56.

- Parametric model: **Gumbel distribution** with unknown location and scale.
- The pseudolikelihood function $\tilde{L}(\pi)$ attains its maximum at:

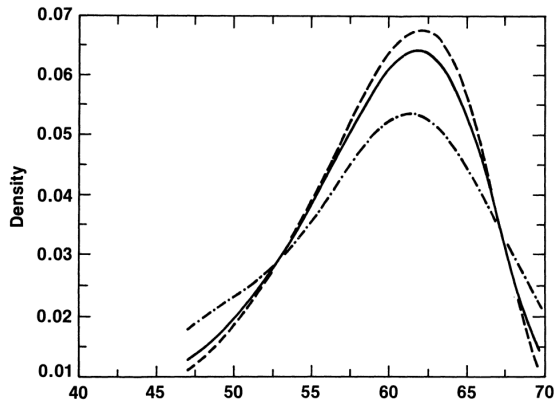
$$\hat{\pi} \approx 0.8.$$

- Interpretation: The parametric model provides a good fit to the data.

Maximum of the Pseudolikelihood ($b = .7s$) for Wind Speed Data.



Density Estimates for Wind Speed Data (case: $b = 0.7s$)

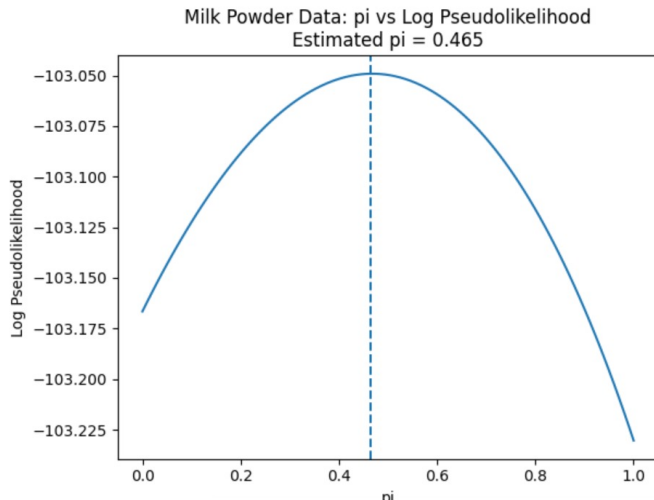


- **Dotted-and-dashed line:** Nonparametric density estimate
- **Dashed line:** Parametric density estimate
- **Solid line:** Semiparametric density estimate

Example 2: Magnesium in Milk Powder

- Data: 22 inductively coupled plasma spectroscopy measurements (micrograms per gram):
1201.1, 1213.3, 1247.6, 1177.1, 1248.9, 1224.1, 1240.5, 1168.4, 1204.4, 1210.1, 1206.5,
1210.0, 1185.6, 1177.7, 1265.7, 1186.4, 1256.7, 1202.3, 1213.7, 1186.3, 1204.9, 1196.7.
- Parametric model: **Normal distribution** with unknown mean and variance.
- As b tends to 0 the nonparametric and semiparametric estimates exhibit **multiple modes**, and this is an indication of difficulty with the measurement process that produced the data.
- Interpretation: Indicates possible complexities or periodic trends in the measurement process.

Maximum of the log Pseudolikelihood ($b = .7s$) for Magnesium in Milk Powder Data.



Behavior of the Semiparametric Estimator

- The semiparametric estimator provides a compromise between parametric and nonparametric approaches.
- Relationship with bandwidth b :
 - $\pi = 0 \Rightarrow$ semiparametric estimate equals the nonparametric estimate with bandwidth $b = 0.5s$.
 - $\pi = 1 \Rightarrow$ semiparametric estimate equals the parametric estimate with bandwidth $b = 0.9s$.
 - As $b \rightarrow 0$, the estimator tends to the nonparametric density.
 - As $b \rightarrow \infty$, it approaches the parametric density.
- This behavior parallels penalized likelihood methods for small and large penalties.

Models and assumption

Model Framework

Suppose that we observe n i.i.d. d -dimensional observations $X_i = (X_{1i}, \dots, X_{di})$, $i = 1, \dots, n$.

- Under an ideal physical model (H_0):

- Observations X_i have density:

$$f(x_i, \theta)$$

- $\theta = (\theta_1, \dots, \theta_k) \in \mathbb{R}^k$
- θ is unknown and estimated from data

- Under less ideal conditions (H_A):

- True density is:

$$f(x_i)$$

- No parametric form assumed
- Must be estimated from the data

- Due to generality of the procedure, several **regularity and rate assumptions** are required

- Parameter θ_0 estimated using MLE:

$$\prod_{i=1}^n f(x_i, \hat{\theta}) = \max_{\theta} \prod_{i=1}^n f(x_i, \theta)$$

- Nonparametric estimator (kernel density):

$$\hat{f}(z) = \frac{1}{nb^d} \sum K\left(\frac{x_i - z}{b}\right)$$

- Semiparametric estimator:

$$g(x, \pi) = \pi f(x, \hat{\theta}) + (1 - \pi) \hat{f}(x), \quad 0 \leq \pi \leq 1$$

- **Consistency and rate of convergence:**

$$n\mathbb{E}\|\hat{\theta} - \theta_0\|^2 = O(1) \quad \text{under } H_0$$

$$n\mathbb{E}\|\hat{\theta} - \theta_A\|^2 = O(1) \quad \text{under } H_A$$

- Interpretation:
 - $\hat{\theta}$ converges, at $\sqrt{n}$ rate, to θ_0 under H_0 and to θ_A under H_A .

Likelihood Ratio Assumptions

Under H_0 ,

$$\sqrt{n} \mathbb{E} \left\{ \frac{\sqrt{f(X_i, \hat{\theta})} - \sqrt{f(X_i, \theta_0)}}{\sqrt{f(X_i, \hat{\theta})}} \right\}^2 = O(1). \quad (a)$$

Under H_A ,

$$\sqrt{n} \mathbb{E} \left\{ \frac{\sqrt{f(X_i, \hat{\theta})} - \sqrt{f(X_i, \theta_A)}}{\sqrt{f(X_i, \hat{\theta})}} \right\}^2 = O(1). \quad (b)$$

As $n \rightarrow \infty$,

$$f(X_i, \hat{\theta}) \xrightarrow{L_2} f(X_i, \theta_0) \quad \text{under } H_0, \quad f(X_i, \hat{\theta}) \xrightarrow{L_2} f(X_i, \theta_A) \quad \text{under } H_A. \quad (c)$$

Assumptions (a) and (b) require the expected likelihood ratios to converge to 1 at the usual rates; (c) requires L_2 convergence of $f(x, \theta)$ to its limit.

In addition, we require conditions on the densities: $f(x)$ and $f(x, \theta)$ have compact support and

$$\inf_x f(x) > \varepsilon, \quad \inf_{x, \theta} f(x, \theta) > \varepsilon,$$

where ε is known. In other words, we require that the underlying density has compact support and is bounded below.

Regularity Conditions

- Smoothness:
 $f(x, \theta)$ is twice continuously differentiable in each argument.
The derivative can be taken under the integral sign.
- Boundedness:

$$\sup_x |f(x)| < M$$

Kernel Density Assumptions

To provide some protection against the possibility that the data come from H_A we use a common nonparametric kernel density estimate

- Kernel estimator:

$$\hat{f}(z) = \frac{1}{nb^d} \sum K\left(\frac{x_i - z}{b}\right)$$

where z is the point at which the density is to be estimated, $K(x)$ is a specified density on R^d and is a continuous kernel with compact support

- Conditions on bandwidth:

$$b \rightarrow 0, \quad nb^{2d} \rightarrow \infty$$

- Ensures:

- Consistency
- Almost sure convergence

- . It has been shown (Devroye 1983) that $\hat{f}(x)$ is an L_1 consistent estimate of $f(x)$.

Non-parametric Rate Assumptions

- Mean square convergence:

$$\mathbb{E} \left[\frac{\hat{f}(X_i) - f(X_i)}{f(X_i)} \right]^2 = O(a_n^2)$$

- Where:

- a_n related to bandwidth b
- $a_n \rightarrow 0$
- $a_n n^{\alpha/2} \rightarrow \infty$ for some α between 0 and 1
- $f(x_i)$ is $f(x_i, \theta_0)$ when H_0 is true and $f(x_i)$ is the true density when H_A is true.

Cross-Product Conditions

- Under H_0 :

$$\int [\hat{f}(x_i) - f(x_i, \theta_0)][\hat{f}(x_j) - f(x_j, \theta_0)] dx_i dx_j = O_p\left(\frac{a_n^2}{n}\right) \quad \text{for } i \neq j$$

- Under H_A :

$$\int [\hat{f}(x_i) - f(x_i)][\hat{f}(x_j) - f(x_j)] dx_i dx_j = O_p\left(\frac{a_n^2}{n}\right) \quad \text{for } i \neq j$$

- Interpretation:
 - Controls dependence between estimation errors

Key Idea of the Model

- Final estimator: convex combination of the parametric and non-parametric estimates

$$g(x, \pi) = \pi f(x, \hat{\theta}) + (1 - \pi) \hat{f}(x)$$

- The parameter π is unknown and its estimate $\hat{\pi}$ is obtained from the data, so that $g(x, \hat{\pi})$ is the actual density estimate.

Let

$$\tilde{L}(\mathbf{x}, \pi) = \prod_{i=1}^n g(x_i, \pi)$$

denote the pseudolikelihood function from which we choose a pseudomaximum likelihood estimate $\hat{\pi}$ for π that satisfies

$$\tilde{L}(\mathbf{x}, \hat{\pi}) = \prod_{i=1}^n g(x_i, \hat{\pi}) = \max_{\pi} \prod_{i=1}^n g(x_i, \pi).$$

- Interpretation: When H_0 holds the chosen density should be close to $f(x, \theta_0)$, and when H_A holds the density should be close to $f(x)$.

Convergence under H_0

- Assume parametric model is true:

$$f(x) = f(x, \theta_0)$$

- Semiparametric estimator:

$$g(x, \pi) = \pi f(x, \hat{\theta}) + (1 - \pi) \hat{f}(x)$$

- Goal:
 - Show $g(x, \hat{\pi})$ behaves like MLE

Theorem:

$$\sqrt{n} \int |g(x, \hat{\pi}) - f(x, \theta_0)| dx = O_p(1)$$

- Achieves parametric convergence rate
- Equivalent to MLE asymptotically

Taylor Expansion Idea

- Consider score function:

$$\frac{\partial \log \tilde{L}(x, \pi)}{\partial \pi}$$

- Expand around:

$$\pi = 1, \quad \theta = \theta_0$$

- First-order terms:
 - Determine convergence of $\hat{\pi}$
- Second-order terms:
 - Asymptotically negligible

- $\log \tilde{L}(x, \pi)$ is a strictly concave function of π with probability 1
- Therefore:
 - $\hat{\pi}$ exists
 - $\hat{\pi}$ is unique (a.s.)
- The validity of the Taylor expansion depends on $\hat{\pi} \neq 0$. It can be proven that

$$\Pr\{\hat{\pi} = 0\} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Expansion Structure

A Taylor expansion of $\frac{\partial \log \tilde{L}(x, \pi)}{\partial \pi}$ about $\pi = 1$ and $\theta = \theta_0$ (evaluated at $\hat{\pi}$) gives:

$$\begin{aligned} 0 = & \sum \frac{f(x_i, \theta_0) - \hat{f}(x_i)}{f(x_i, \theta_0)} \\ & - (\pi - 1) \sum \frac{(f(x_i, \theta_0) - \hat{f}(x_i))^2}{f(x_i, \theta_0)} \\ & - \sum \frac{f(x_i, \theta_0) - \hat{f}(x_i)}{f(x_i, \theta_0)} (\hat{\theta} - \theta_0) \left(\frac{\nabla_{\theta} f(x_i, \theta_0)}{f(x_i, \theta_0)} \right)^T \\ & + (\hat{\theta} - \theta_0) \sum \left(\frac{\nabla_{\theta} f(x_i, \theta_0)}{f(x_i, \theta_0)} \right)^T + \text{higher-order terms} \end{aligned}$$

- First-order term:
 - Controls convergence rate
- Higher-order terms:
 - $o_p(1)$

Result for $\hat{\pi}$

- From expansion it can be shown:

$$\hat{\pi} - 1 = O_p\left(\frac{1}{\sqrt{na_n}}\right)$$

- Therefore:

$$\hat{\pi} \rightarrow 1$$

- Parametric model dominates

- Writing $g(x, \hat{\pi}) - f(x, \theta_0) = \hat{\pi} [f(x, \hat{\theta}) - f(x, \theta_0)] + (1 - \hat{\pi}) [\hat{f}(x) - f(x, \theta_0)]$. and applying triangle inequality on $\int |g(x, \hat{\pi}) - f(x, \theta_0)| dx$ we get:

$$\int |g(x, \hat{\pi}) - f(x, \theta_0)| dx = O_p(n^{-1/2})$$

- **Conclusion:**
 - Same behavior as parametric estimator

Convergence under H_A

- True model is nonparametric:

$$f(x) \neq f(x, \theta)$$

- Goal:
 - Show estimator behaves like kernel estimator

Theorem 2. Under the assumptions stated previously

$$\left| \frac{f(x_i, \theta_A) - f(x_i, \hat{\theta})}{f(x_i, \theta_A)} \right| \leq |\hat{\theta} - \theta_A| R(x_i),$$

where $E[R(X_i)]^2 < \infty$. Then

$$a_n^{-1} \int |g(x, \hat{\pi}) - f(x)| dx = O_p(1). \quad (\text{d})$$

Proof. To prove (d), we show that

$$\frac{1}{n} \sum \log g(x_i, \pi) = -O \left(\int \left| \sqrt{g_A(x, \pi)} - \sqrt{f(x)} \right|^2 dx \right) + O_p(a_n^2), \quad (\text{e})$$

where

$$g_A(x, \pi) = \pi f(x, \theta_A) + (1 - \pi)f(x).$$

- Result (e) is stronger than (d)
- $\hat{\pi}$ maximizes (e)
- Equivalent to minimizing Hellinger distance

Step 1

- Show:

$$\frac{1}{n} \sum \log g(x_i, \pi) = \mathbb{E}[\log g_A(X, \pi)] + O_p(n^{-1/2})$$

- Method:

- Taylor expansion around θ_A

Step 2

- Show:

$$\mathbb{E}[\log g_A(X, \pi)] = -O\left(\int (\sqrt{g(x, \pi)} - \sqrt{f(x)})^2 dx\right)$$

- Interpretation:

- Hellinger distance

Step 1 and **Step 2** together imply that

$$\hat{\pi} = O_p(n^{-1/4}).$$

By iterating this process we ultimately achieve (e).

- Using the expansion of $H(\pi) = \frac{1}{\sqrt{n}} \sum [\log g_A(x_i, \pi) - E \log g_A(X, \pi)]$ about $\pi = 0$ and using an iterating argument from previous calculations (from which we know that $\hat{\pi} = O_p(\max(a_n, n^{-1/2}))$), it can be shown $\hat{\pi} = O_p(a_n)$ provided that $a_n n^{\alpha/2} \rightarrow \infty$ for some $0 < \alpha < 1$.
- Hence, Non-parametric model dominates.

Clustering

What does clustering stand for?

Clustering

The clustering process aims at grouping together points having high similarity, placing those with large dissimilarity into different groups.

Various Types of Clustering

- Traditional Approaches
 - Partition-based
 - Distance-based
 - Grid-based
- Model-based
- Density-based

Model-Based Clustering

In model-based clustering, a finite mixture of parametric distributions is introduced. More precisely, the random vector X is considered to be a mixture of J sub-populations.

Density Model

$$f(x) = \sum_{j=1}^J \pi_j f_j(x)$$

where f_j denotes the density of the j -th sub-population and π_j represents its relative weight.

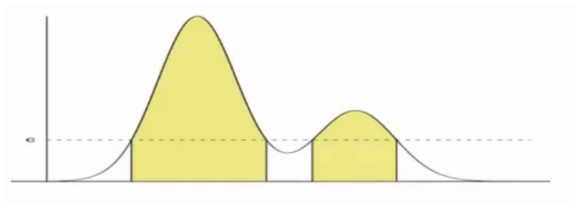
In this framework, a cluster is associated with each component f_j , and any observation x_0 is allocated to the cluster with the maximal density $f_j(x_0)$ among the J components.

Limitations

- To render the estimation problem identifiable, some restrictions must be placed on the components.
- The number of components should be determined in advance.
- If the cluster shapes do not match those of the components f_j , the model-based approach may face difficulties.

Density-Based Clustering

The mechanism behind density-based clustering is to associate groups with **connected regions of the density**.



Key Idea

- Clusters correspond to **high-density regions (modes)**
- Separation occurs in **low-density areas**

Kernel Density Estimation (KDE)

Suppose that $X_1, \dots, X_n$ is a random sample from a density f .

KDE Formula

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} \kappa\left(\frac{x - X_i}{h}\right)$$

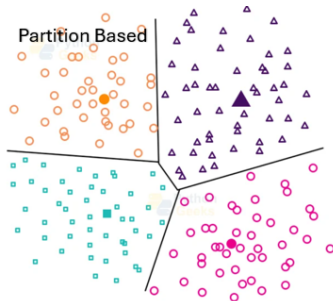
Components

- $\kappa(\cdot)$: kernel function
- h : bandwidth (controls smoothness)

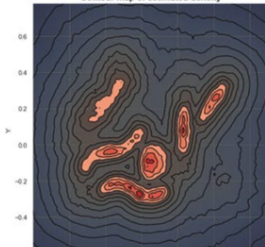
The Non-Parametric pdfCluster

- **Azzalini and Torelli (2007):** Proposed a *density-based clustering* procedure based on *Kernel Density Estimation (KDE)*. The method exploits *spatial tessellation* to generalize the concept of *adjacency* in multidimensional spaces and to identify *connected level sets*. Clusters are defined as *maximally connected components* of regions where the estimated density exceeds a given threshold.
- **Limitation:** The algorithm is computationally feasible only for *low to moderate dimensional data* (typically up to dimension 6).
- **Menardi and Azzalini (2014):** Extended the methodology by providing an improved approach for detecting *connected high-density regions* in *higher-dimensional spaces*.
- **Implementation:** Both methods are unified into a single algorithm and implemented in the R package pdfCluster (Azzalini and Menardi, 2014).

Types of Clustering Approach



Contour map of estimated density



Distance-based Clustering

- Select $X = k$ points in $\mathbb{R}^d$, that minimize

- K-Medians

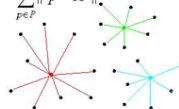
$$\sum_{p \in P} \|p - X\|$$

- Each point is assigned to closest center in X

- Cluster: Set of points assigned to a center

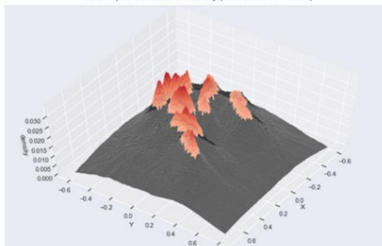
- K-Means

$$\sum_{p \in P} \|p - X\|^2$$



Facility Location

Surface plot of estimated density (core distance < 0.045)



Why Asymmetric Kernel?

- Classical kernel methods are based on symmetric densities, typically centered around zero, as in the case of the normal kernel.
- However, a deficiency arises when such kernels are used to estimate densities whose supports are bounded [*Marron and Ruppert (1994)*].
- On the other hand, asymmetric kernel methods never assign weight outside the support of the density. As a result, they effectively address the boundary bias problem [*Saulo et al., 2013*].
- Density estimators in moderately-sized samples tend to inherit the salient properties of their kernels. Thus, if a density is suspected to be highly skewed, it is more appropriate to use asymmetric kernels rather than symmetric ones [*Abadir and Lawford (2004)*].

Proposed Approach: The Semi-Parametric pdfcluster

- Our main goal in this work is to improve the density estimation component of the pdfCluster algorithm [Azzalini and Menardi (2014)] by utilizing skew-symmetric kernels instead of the normal and t_7 kernels.
- As the terminology suggests, skew-symmetric (or skew-modulated) distributions are capable of generating both symmetric and asymmetric densities; see, for example, [Salehi and Azzalini (2018)].
- Among the many distributions in this family, the skew-normal and skew- t distributions—being the most well-known members—are sufficiently flexible to be employed as kernels in kernel density estimation (KDE).

Semi-Parametric KDE

Semi-Parametric KDE

Suppose that

$$\mathbf{X} := (X_1, \dots, X_d), \quad i = 1, \dots, n,$$

is a random sample of size n from a d -dimensional density function, and let

$$\mathbf{X}_i = (X_{i1}, \dots, X_{id})$$

be an observation.

Azzalini and Menardi (2014) performed the density estimation step of their clustering algorithm using a kernel method based on a product kernel estimator of the form:

$$\hat{f}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \prod_{j=1}^d \frac{1}{h_{ij}} \kappa\left(\frac{x_j - X_{ij}}{h_{ij}}\right), \quad (8)$$

where

$$\mathbf{x} = (x_1, \dots, x_d),$$

$\kappa(\cdot)$ is a symmetric kernel (e.g., normal or t_7 as used in `pdfCluster`), and h_{ij} denotes the bandwidth corresponding to X_{ij} for $i = 1, \dots, n$ and $j = 1, \dots, d$.

Proposed Skew-Symmetric Kernel Estimator

Here, we intend to employ a product kernel of the following form:

$$\hat{f}(\mathbf{x}) = \frac{1}{n h_1 \cdots h_d} \sum_{i=1}^n \prod_{j=1}^d \kappa_j \left(\frac{x_j - X_{ij}}{h_j} \right), \quad (9)$$

where $\kappa_j(\cdot)$ can be either a skew-normal density or a skew- t density. More precisely, in the skew-normal case, we have:

$$\kappa_j(z) = 2 \phi(z) \Phi(\lambda_j z), \quad z \in \mathbb{R}, \quad (10)$$

where λ_j , $j = 1, \dots, d$, is the slant parameter, and $\phi(\cdot)$ and $\Phi(\cdot)$ denote the density and cumulative distribution function of the standard normal distribution, respectively.

Proposed Skew-Symmetric Kernel Estimator(Contd.)

Our second proposal for $\kappa_j(\cdot)$ is the standard skew- t density given by:

$$\kappa_j(z) = 2 t(z; \nu_j) T\left(\lambda_j z \sqrt{\frac{\nu_j + 1}{\nu_j + z^2}}; \nu_j + 1\right), \quad z \in \mathbb{R}, \quad (11)$$

where λ_j is the slant parameter and ν_j , $j = 1, \dots, d$, denotes the degrees of freedom. Here, $t(\cdot; \nu)$ and $T(\cdot; \nu)$ represent the density and cumulative distribution function of the classical Student's t distribution with ν degrees of freedom, respectively.

Our Algorithm

Let

$$\mathbf{z}_j := (Z_{1j}, \dots, Z_{nj}).$$

- Choose the desired kernel between (3) and (4).
- For the skew-normal (SN) kernel, plug in λ_j using its MPLE, assuming $\mathbf{z}_j$ is a random sample from (5). (*A point on MPLE*)
- For the skew- t (ST) kernel, proceed as follows, assuming $\mathbf{z}_j$ is a random sample from (6):
 - (i) For high dimensions: approximate (λ_j, ν_j) with quick preliminary estimates using `st.prelimFit`.
 - (ii) For low dimensions: plug in (λ_j, ν_j) using their MPLEs via the routines `st.prelimFit` and `st.mple`.
- Compute $\hat{f}(\mathbf{x})$ in (4) with $\kappa_j(\cdot)$ whose parameters are now approximated.
- The only remaining ingredient is to determine an optimal bandwidth, which will be discussed in the next section.

On finding optimum bandwidth

Suppose that $f(\cdot)$ is a univariate density function to be estimated using (2), considering $d = 1$. Assume further that $\kappa(\cdot)$ is a one-dimensional asymmetric kernel density satisfying the following regularity assumptions:

Regularity Assumptions

$$(A_1) \quad z \in \mathbb{R}, \quad \kappa(z) \geq 0$$

$$(A_2) \quad \int \kappa(z) dz = 1$$

$$(A_3) \quad \int z \kappa(z) dz = \mu_1(\kappa) \neq 0$$

Optimal Bandwidth h_{opt}

Now, if we consider $\kappa(\cdot)$ to be the one given by (3), then h_{opt} is obtained as a real value satisfying:

Equation

$$-a + 2b h_{\text{opt}}^3(\lambda) + c h_{\text{opt}}^5(\lambda) = 0 \quad (5)$$

where,

$$a := \frac{1}{n} \left(2\pi^{\frac{-3}{2}} \arctan\left(\sqrt{1 + \lambda^2}\right) \right)$$

$$b := \frac{\lambda^2}{2\pi^{3/2} \sigma^3 (1 + \lambda^2)}$$

$$c := \frac{3}{8\pi^{1/2} \sigma^5} \quad (6)$$

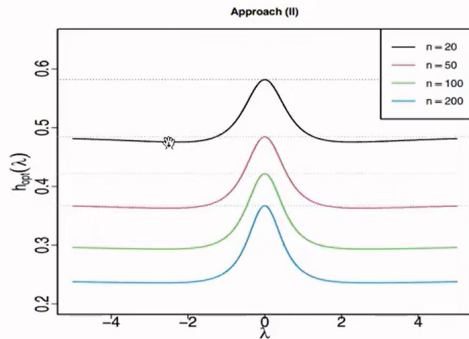


Figure 1: The plot of h_{opt} versus λ for various sample sizes and $\sigma = 1$. The dotted lines correspond to the optimum bandwidths obtained under the normality assumption.

While this strategy may appear naive, it does produce sensible results in virtually all cases on which we have tested the method; this conclusion is based on the examples reported here plus some additional simulated data. Empirically, it was observed that it is often advantageous to shrink h slightly towards zero; in the subsequent numerical work, we adopted a shrinkage factor of $3/4$ as an overall reasonable choice.

Multi-dimensional Case

Assume $f(\mathbf{x})$ is a d -variate density. Then, the kernel density estimator in its most general form is defined as

[Wand and Jones (1995)]:

Kernel Density Estimator

$$\hat{f}(\mathbf{x}) = \frac{1}{n} |\mathbf{H}|^{-1/2} \sum_{i=1}^n K\left(\mathbf{H}^{-1/2}(\mathbf{x} - \mathbf{x}_i)\right) \quad (7)$$

Notation

- $\mathbf{x} = (x_1, \dots, x_d)$
- $\mathbf{X}_i = (X_{i1}, \dots, X_{id}), \quad i = 1, \dots, n$
- $\mathbf{H}$: bandwidth matrix (symmetric positive definite $d \times d$ matrix)
- $K(\cdot)$: d -variate kernel density

Theorem: AMISE in Multi-dimensional Case

Theorem

Assume $h_j = h$, for $j = 1, \dots, d$. Then, the AMISE of $\hat{f}(\mathbf{x})$ has the form:

$$\text{AMISE}(\hat{f}) = \frac{1}{nh^d} \int t^{1/3}(\mathbf{x}) d\mathbf{x} + A (Bh^2 + Ch^4) \quad (8)$$

where,

$$B := 8 \sum_{j=1}^d \mu_1^2(\kappa_j) \sigma_j^{-2}$$

$$A := (2\sqrt{\pi})^{-d} \prod_{j=1}^d \sigma_j^{-1}$$

$$C := 3 \sum_{j=1}^d \mu_2^2(\kappa_j) \sigma_j^{-4} + 2 \sum_{j < l} [\mu_2(\kappa_j) \mu_2(\kappa_l) + \mu_1^2(\kappa_j) \mu_1^2(\kappa_l)] \sigma_j^{-2} \sigma_l^{-2}$$

Minimizing (8) gives the optimum bandwidth as the solution of the following non-linear equation

$$-\frac{d}{n} \prod_{j=1}^d \int \kappa_j^2(t) dt + 2ABh_{\text{opt}}^{d+2} + 4ACH_{\text{opt}}^{d+4} = 0, \quad (9)$$

where constants A , B and C are given by (9). The unknown parameters σ_j can be substituted by s_j , the sample standard deviation of the j th variable, and the λ_j can be approximated by its MPLE.

It can be easily shown that by setting $d = 1$ in (9), it simplifies to its univariate version given by (5).

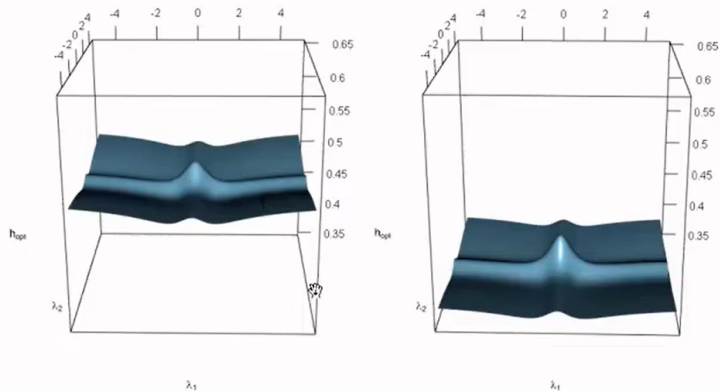


Figure 2: The plot of h_{opt} versus the pair of (λ_1, λ_2) for $\sigma_1 = \sigma_2 = 1$, $n = 50$ (the left panel) and $n = 200$ (the right panel).

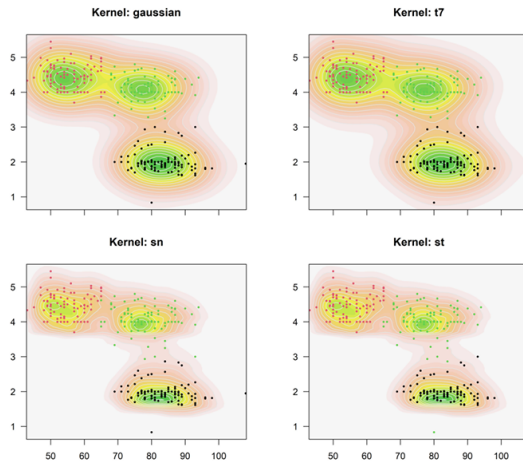


Figure 7: Clustering results of *geyser* data as well as contour levels of the various fitted kernel estimates.

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Thank You!

Questions?